MIDDLESEX SCHOOL | MATHEMATICS

Math 31 Course at a Glance

Advanced Algebra

Review Content

REVIEW	
M31-REV-001	I can calculate a percentage, percent increase, or percent decrease.
M31-REV-002	I can simplify an expression using exponent properties.
M31-REV-003	I can use polynomial vocabulary to distinguish terms, factors, coefficients, degree, and polynomial structure.
M31-REV-004	I can expand a product of two binomials.

Linear Functions and Function Notation

REQUIRED	
M31-LIN-001	I can interpret function notation as a statement about input and output.
M31-LIN-002	I can determine whether a graph or table represents a function.
M31-LIN-003	I can evaluate a function or solve for an input using a graph, table, or formula.
M31-LIN-004	I can identify independent and dependent variables in a relationship.
M31-LIN-005	I can calculate average rate of change from two points.
M31-LIN-006	I can interpret interval notation and identify increasing or decreasing intervals.
M31-LIN-007	I can determine total change from an average rate of change over an interval.
M31-LIN-008	I can determine and interpret the units of an average rate of change.
M31-LIN-009	I can identify a linear function from a table, graph, or formula.
M31-LIN-010	I can interpret the slope and intercept of a linear model.
M31-LIN-011	I can determine the intersection of two linear relationships graphically or algebraically.
M31-LIN-012	I can write equations of horizontal, vertical, parallel, and perpendicular lines through a specified point.

Function Behavior and Structure

REQUIRED	
M31-FUN-001	I can determine whether a relation is a function.
M31-FUN-002	I can evaluate a function from a formula, graph, or table.
M31-FUN-003	I can determine domain and range from a formula, graph, or table.
M31-FUN-004	I can determine algebraic domain restrictions caused by denominators and even roots.
M31-FUN-005	I can graph a piecewise function from its formula.
M31-FUN-006	I can write a piecewise formula that represents a graph.
M31-FUN-007	I can graph a transformed function using shifts, stretches, compressions, and reflections.
M31-FUN-008	I can describe how a transformation affects domain, range, or asymptotes.
M31-FUN-009	I can evaluate a composition from formulas, graphs, or tables.
M31-FUN-010	I can write a formula for a composition of two functions.
M31-FUN-011	I can find an inverse function algebraically.
M31-FUN-012	I can evaluate and interpret an inverse from a graph, formula, or table.
M31-FUN-013	I can determine and interpret the units of an inverse function.
M31-FUN-014	I can explain how the domain and range of a function relate to those of its inverse.
M31-FUN-015	I can determine the range of a function by analyzing the domain of its inverse.
M31-FUN-016	I can verify algebraically that two functions are inverses.
M31-FUN-017	I can interpret the meaning and units of a composition.
M31-FUN-018	I can classify a graph or table as concave up or concave down using changes in average rate of change.
M31-FUN-019	I can determine whether a function is invertible using the horizontal line test.
M31-FUN-020	I can graph an inverse function as a reflection across the line `y = x`.

Quadratic Functions and Equations

REQUIRED	
M31-QUAD-001	I can identify the vertex, intercepts, axis of symmetry, and concavity of a quadratic function.
M31-QUAD-002	I can connect standard, factored, and vertex forms to features of a quadratic graph.
M31-QUAD-003	I can rewrite a quadratic function in a form that reveals a requested feature.
M31-QUAD-004	I can solve a quadratic equation by factoring or using the quadratic formula.
M31-QUAD-005	I can graph a quadratic function using its algebraic structure and key features.
M31-QUAD-006	I can construct a quadratic function from specified zeros, a vertex, or graphical information.
M31-QUAD-007	I can create and interpret a quadratic trajectory model.

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Exponential Functions and Models

REQUIRED

M31-EXP-001	I can distinguish linear change from exponential change using differences, ratios, or percent change.
M31-EXP-002	I can identify initial value and growth or decay factor in an exponential model.
M31-EXP-003	I can write an exponential function from a table, graph, or context.
M31-EXP-004	I can evaluate and interpret an exponential model.
M31-EXP-005	I can calculate growth under periodic compounding.
M31-EXP-006	I can calculate growth under continuous compounding.
M31-EXP-007	I can determine an exponential formula from two points or other sufficient data.
M31-EXP-008	I can determine domain, range, intercepts, and asymptotes of an exponential function.
M31-EXP-009	I can compare the growth rates of two exponential functions.
M31-EXP-010	I can determine the percent growth or decay represented by a continuous exponential model.

Logarithmic Functions and Models

REQUIRED

M31-LOG-001	I can convert between logarithmic and exponential forms.
M31-LOG-002	I can evaluate a logarithm using its exponential meaning.
M31-LOG-003	I can expand or condense an expression using product, quotient, and power properties.
M31-LOG-004	I can solve an exponential equation using logarithms.
M31-LOG-005	I can solve a logarithmic equation using exponential form and check domain restrictions.
M31-LOG-006	I can calculate and interpret doubling time or half-life.
M31-LOG-007	I can determine domain, range, intercepts, and asymptotes of a logarithmic function.
M31-LOG-008	I can explain how exponential and logarithmic functions undo one another.
M31-LOG-009	I can solve a multi-step exponential or logarithmic equation.
M31-LOG-010	I can determine an intersection of exponential functions graphically or algebraically.
M31-LOG-011	I can rewrite a discrete exponential model as an equivalent continuous exponential model.
M31-LOG-012	I can solve and interpret an exponential or logarithmic modeling problem.

Extension Content

EXTENSION

M31-EXT-001	I can analyze an absolute-value function using transformations.
M31-EXT-002	I can determine even or odd symmetry from a graph or formula.
M31-EXT-003	I can interpret a contextual logarithmic scale.